4.1

Production of Goods and Services

Cambridge IGCSE Business

- Labour Productivity
- Efficiency
- Lean Production
- JIT & Kaizen
- Methods of Production
- Inventory

Operations Management

Operations management covers how a business makes its products — efficiently, at low cost and with minimum waste.

- Calculate and improve labour productivity
- Understand lean production and waste reduction
- Apply JIT inventory and Kaizen methods
- Compare job, batch and flow production
- Manage inventory effectively

Labour Productivity & Efficiency

Measuring output · Why efficiency matters · How to improve it

Labour Productivity – Definition & Formula

Labour productivity measures how much output each worker produces in a given time period. It is a key indicator of operational efficiency.

Labour Productivity Formula

$$\text{Labour Productivity} = \frac{\text{Output per period (units)}}{\text{Number of employees}}$$

Worked Example

A factory produces 6,000 units per week with 30 workers.

Labour Productivity = 6,000 ÷ 30 = 200 units per worker per week

How to improve labour productivity

- Increase automation — machines replace repetitive tasks
- Better training — skilled workers produce more per hour
- Improve processes — remove bottlenecks and wasted steps
- Financial incentives — bonuses and piece-rate pay motivate workers

Amazon

750,000 robots in warehouses

increase speed, cut labour costs

Why productivity matters

- Lower cost per unit — more competitive prices
- Higher output — more revenue
- Signals efficient use of resources to investors
- Enables business growth without proportional cost rise

Lean Production

Eliminating waste · JIT · Kaizen · Toyota philosophy

Lean Production – What It Is & The 7 Wastes (TIMWOOD)

Lean production is a philosophy that aims to maximise value by eliminating all forms of waste — anything that doesn't add value for the customer.

Lean Production — Eliminating 7 Types of Waste (TIMWOOD)

T	I	M	W	O	O	D
Transport	Inventory	Motion	Waiting	Overproduction	Over-processing	Defects
Moving materials unnecessarily	Holding too much stock	Unnecessary staff movement	Idle time in production	Making more than needed	Doing more than required	Faults requiring rework or scrap

Two main lean methods you must know:

Just-In-Time (JIT) Inventory

- Stock is ordered and delivered exactly when needed — none held in advance
- Eliminates storage costs and reduces waste from obsolete stock
- Requires very reliable suppliers and precise delivery schedules
- EXAMPLE: Toyota receives parts from suppliers just hours before assembly

Kaizen (Continuous Improvement)

- Kaizen = Japanese for 'change for the better'
- All workers — not just managers — suggest small improvements daily
- Small changes accumulate into significant efficiency gains over time
- EXAMPLE: Toyota workers stopped the production line 1,000 times per day to fix problems

Exam tip: Toyota is the best example for both JIT and Kaizen — they invented both methods. Always link to the TIMWOOD 7 wastes.

Lean Production – JIT in Detail & Advantages vs Disadvantages

Just-In-Time inventory control means stock arrives exactly when it is needed in production — no earlier, no later.

Just-In-Time (JIT) — How It Works

Supplier
Delivers parts exactly when needed — no sooner

→

Factory
No stock warehouse needed
Parts go straight to production

→

Product
Completed just in time for customer order

→

Saving
Zero storage costs
Minimum waste

→

Toyota
Pioneer of Just-In-Time (JIT) production — zero waste philosophy

JIT Advantages

- No storage costs — warehouse not needed
- Reduces cash tied up in unsold stock
- Less waste — perishable stock does not expire
- Faster response to changes in customer demand
- Lower risk of stock becoming obsolete

JIT Disadvantages

- Very dependent on reliable, on-time suppliers
- No buffer if demand suddenly spikes
- Production stops if delivery is late or stock runs out
- Requires close coordination across supply chain
- Higher individual delivery costs per order

Inventory Management – Why Businesses Hold Stock

Inventory (stock) is the raw materials, work-in-progress and finished goods a business holds. Managing inventory is a balancing act — too much wastes money, too little causes problems.

Why businesses hold inventory:

- Meet customer demand quickly without delays
- Keep production running if deliveries are late
- Buy in bulk to get lower prices from suppliers
- Buffer against supply disruptions or supplier failures
- Seasonal businesses stock up before peak demand

Factors affecting how much stock to hold:

- How reliable are suppliers? Unreliable = hold more
- How variable is demand? High variation = hold more
- Storage cost — expensive to store = hold less
- Lead time — how long does delivery take?

Raw Materials

Steel, cotton, flour — inputs waiting to be used in production

Work-in-Progress

Half-finished goods currently being processed or assembled

Finished Goods

Completed products waiting to be sold to customers

Methods of Production

Job · Batch · Flow

Methods of Production – Job, Batch and Flow

The method of production a business chooses depends on the type of product, the size of the market and the level of customisation required.

Job Production

JOB PRODUCTION

One product at a time

Made to individual customer spec

e.g. wedding cakes · bridges

Rolls-Royce · Taylor-made suits

Each product is made individually, to a specific customer order.

Rolls-Royce cars · wedding cakes · tailored suits · one-off bridges

- ✅ Unique, customised products add high value
- ✅ Skilled workforce — workers are motivated
- ✅ Premium prices can be charged
- ❌ Very high labour costs per unit
- ❌ Slow — each item takes a long time
- ❌ Difficult to scale up production quickly

Batch Production

BATCH PRODUCTION

Groups of identical products

One batch finished before next starts

e.g. 200 white loaves, then 200 brown

Bakeries · clothing manufacturers

Groups of identical products are made together. One batch is completed before the next starts.

Bakeries (white loaves, then brown) · clothing (S, M, L sizes) · printing

- ✅ More flexible than flow — batches can change
- ✅ Lower cost per unit than job production
- ✅ Can produce a range of different products
- ❌ Machine set-up time wasted between batches
- ❌ Work-in-progress stock ties up capital
- ❌ Less customised than job production

Flow Production

FLOW PRODUCTION

Continuous non-stop production

Same product 24 hours a day

e.g. Coca-Cola bottles · car assembly

McDonald's · oil refineries

Identical products made continuously on a production line, 24 hours a day.

Coca-Cola bottles · car assembly (Ford) · oil refining · smartphone chips

- ✅ Very low cost per unit — economies of scale
- ✅ High output — thousands of units per hour
- ✅ Consistent, standardised quality
- ❌ Very high set-up costs for machinery
- ❌ Inflexible — hard to change product quickly
- ❌ Repetitive work — workers can become demotivated

Exam tip: The method chosen links to pricing, workforce skills, equipment costs and the marketing mix. Always justify your recommendation.

Key Revision Checklist – 4.1 Production of Goods and Services

Labour Prod.	Formula = Output ÷ Number of employees. How to improve: automation, training, better processes, incentives
Efficiency	Why important: lower unit costs, higher competitiveness, better use of resources. Always link to business objectives
Lean Prod.	Eliminating waste — TIMWOOD (Transport, Inventory, Motion, Waiting, Overproduction, Over-processing, Defects)
JIT	Stock delivered exactly when needed — advantages: no storage costs. Disadvantages: relies on suppliers, no buffer stock
Kaizen	Continuous improvement — all workers suggest small daily improvements. Toyota example: 1,000 production line stops per day to fix issues
Inventory	Why hold stock: meet demand, buffer against delays, bulk buying. Types: raw materials, WIP, finished goods
Job Prod.	One at a time, customised, high cost, high quality. Examples: Rolls-Royce, wedding cakes, tailored suits
Batch Prod.	Groups of identical items. Flexible but set-up time wasted between batches. Examples: bakeries, clothing
Flow Prod.	Continuous, standardised, low cost, high volume. Examples: Coca-Cola, car assembly lines, oil refining